

GHIBRAN ESA ANUGRAH

Telecommunications Engineering | Politeknik Negeri Semarang |
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PROFESSIONAL SUMMARY

Telecommunications Engineering graduate from Politeknik Negeri Semarang with hands-on industry experience at PT Telkom Akses in fiber optic and FTTx network operations. Certified Radio Frequency Engineer with CCNA certifications, supported by practical projects in RF engineering, wireless communications, digital signal processing, embedded systems, and IoT. Proficient in network troubleshooting, RF and antenna design, system development, and technical analysis, with the ability to integrate hardware, communication, and software components into practical engineering solutions.

EDUCATION

POLITEKNIK NEGERI SEMARANG

Diploma III in Telecommunications Engineering
Tembalang, Semarang, Indonesia
GPA : 3.62 / 4.00

September 2023 – September 2026

SMK NEGERI 1 SLAWI

Computer and Network Engineering
Slawi, Tegal, Indonesia

July 2020 – April 2023

WORK EXPERIENCE

PT TELKOM AKSES

Intern Staff | September 2025 – December 2025
Tegal, Indonesia

- Executed emergency restoration and core-to-core fusion splicing for a 144-core fiber optic feeder cable severed during utility excavation along Jl. Gajah Mada, Tegal.
- Conducted fault location, link loss measurement, and post-repair attenuation verification across reconstructed spans using Optical Time-Domain Reflectometer (OTDR).
- Maintained OLT, ODC, and ODP infrastructure to ensure continuous network stability while documenting technical restoration logs per Telkom Akses standards.

PT TELKOM AKSES

Intern Staff | June 2022 – September 2022
Slawi, Indonesia

- Provisioned and configured Optical Network Terminals (ONT) and Set-Top Boxes (STB) for new IndiHome and IndiBiz subscribers.
- Performed drop-cable deployments, Optical Power Meter (OPM) signal measurements, and validated parameters for IPTV, VoIP, and broadband services.
- Resolved customer-side network incidents (onsite troubleshooting), replacing damaged devices and ensuring SLA installation standards were met.

PROJECT EXPERIENCE

IoT-Based Smart Camera Cabinet with Automated Hysteresis Climate Control

Internet of Things (IoT) & Embedded Systems | 2026

- Built an ESP32-powered camera storage cabinet using a Peltier module, featuring automated humidity regulation, UVC sterilization, and real-time web monitoring.
- Achieved up to 99.21% humidity control accuracy with stable continuous operation exceeding 144 hours.

Microstrip Antenna Design Projects

RF Engineering / Antenna Design | 2025

- Designed and simulated single, 2x2, and 4x1 microstrip antenna configurations for the 902–928 MHz ISM band using CST Studio Suite.
- Evaluated S11, VSWR, gain, directivity, and radiation patterns for sub-GHz applications, achieving optimal S11 around -28 dB and VSWR of 1.1–1.3.

OFDM Wireless Communication Simulator

Wireless Communications / Digital Signal Processing | 2025

- Developed an end-to-end OFDM communication simulator evaluating QPSK, 16-QAM, and 64-QAM modulations over AWGN and Rayleigh channel models.
- Executed FFT/IFFT processing, Zero-Forcing equalization, and BER analysis to evaluate wireless transmission performance across varied SNR conditions.

Radar Target Detection Simulator via Matched Filtering

Digital Signal Processing / Radar Systems | 2025

- Developed a radar signal processing simulator integrating Linear Frequency Modulation (LFM) chirp generation, multi-target echo modeling, AWGN interference, and matched filtering.
- Implemented peak detection algorithms for target range estimation, successfully identifying three simulated targets with zero range estimation error.

ECG Heart Rate Detection & Arrhythmia Analysis System

Biomedical Electronics / Signal Processing | 2026

- Developed an ECG signal processing tool in GNU Octave to filter noise and process raw physiological heartbeat data from the MIT-BIH database.
- Implemented noise filtering and adaptive peak detection to estimate heart rate metrics and classify basic arrhythmia conditions.

Water Resource Monitoring & Billing Management System

Web Application / Smart Water Management System | 2025

- Built a web-based water management application using PHP and MySQL to automate meter recording and customer billing.
- Integrated water usage analytics, customer data management, tariff calculation schemes, and payment history tracking.

CERTIFICATIONS

Radio Frequency Engineer - 193107501110-44/VSGA/BLSDM.Komdigi/2025

Vocational School Graduate Academy (VSGA), Digital Talent Scholarship 2025

Balai Pelatihan Sumber Daya Manusia Komunikasi dan Digital (BLSDM Komdigi) Yogyakarta

CCNA: Switching, Routing and Wireless Essentials

Cisco Networking Academy

CCNA: Introduction to Networks

Cisco Networking Academy

KKNI Level II – Computer and Network Engineering Competency - 61100 3513 2 0000258 2023

Badan Nasional Sertifikasi Profesi (BNSP)

ORGANIZATIONAL EXPERIENCE

KELUARGA MAHASISWA BIDIKMISI POLITEKNIK NEGERI SEMARANG

President | September 2024 – September 2025

- Directed organizational strategy and managed end-to-end execution of academic and leadership programs, building strategic partnerships with internal and external stakeholders.
- Led executive committee members, overseeing budget allocation, project schedules, and performance evaluations for student development initiatives.

HIMPUNAN MAHASISWA ISLAM KOMISARIAT POLINES

General Secretary | June 2025 – Present

- Managed central administrative operations, optimizing official documentation, archive systems, and meeting minutes to ensure compliance.
- Coordinated cross-functional communication among executive members and structured periodic operational reports to enhance organizational transparency.

AWARDS & ACHIEVEMENTS

- 1st Place – Sultan Agung Literacy Fest Essay Competition 2025, Badan Eksekutif Mahasiswa Universitas Sultan Agung
- 2nd Place – Information Network Cabling, Tegal Regency Student Competency Competition 2022, Musyawarah Kerja Kepala Sekolah SMK Kabupaten Tegal
- Silver Medal & Bronze Medal – Pekan Essay Nasional (PENA) 2, Universitas Dhyana Pura
- Best Paper – National Essay Competition 2025, Universitas Negeri Semarang Scholarship Community
- 2nd Runner-Up – Fair Essay Competition 2025, Forum Komunikasi Mahasiswa Politeknik Indonesia, Politeknik Negeri Semarang

TECHNICAL SKILLS

Telecommunications & Networks	: RF & Cellular Networks, Fiber Optic & FTTx, Computer Networking, Routing & Switching, Wireless Networking, Network Troubleshooting
Embedded & Hardware	: ESP32, Arduino, C/C++, IoT Systems, Sensors & Actuators, HTTP/API Communication, MQTT, System Integration
RF & Signal Processing	: RF Engineering, Digital Signal Processing, OFDM, Radar Signal Processing, Matched Filtering, Microstrip Antenna Design
Software & Tools	: Cisco, MikroTik, Cacti, PRTG, Wireshark, Nmap, Atoll, Pathloss, TEMS Investigation, OptiSystem, GNU Octave, Proteus, CST Studio Suite, Microsoft Office
Web & Database	: PHP, JavaScript, HTML/CSS, MySQL

PROFESSIONAL SKILLS

Technical & Operational	: Problem Solving & Technical Troubleshooting, Technical Documentation, Field Operation & Maintenance, System Integration
Management & Soft Skills	: Leadership & Team Coordination, Project Management, Analytical Thinking, Communication & Stakeholder Coordination

LANGUAGES

Indonesian	: Native
English	: Intermediate (POLINES TEPPS Score: 660 Listening: 400, Reading: 260)